

Rahul Anand Nayanegali

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SUMMARY

Software Engineer with 6 years of experience building full-stack and backend systems across Amazon Seller Central, AI-powered data tools, and high-traffic streaming platforms. Strong in Java, TypeScript, Node.js, Spring Boot, PostgreSQL, REST/GraphQL APIs, AWS, and production-quality testing.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript

Backend: Java/Spring Boot, Node.js, Flask, REST APIs, GraphQL, gRPC, WebSockets, SSE

Databases: PostgreSQL, SQL

Cloud/Infra: AWS ECS, EC2, Lambda, S3, CloudFront, Docker, CI/CD

Frontend: React, Next.js, TypeScript

Testing: TDD, BDD, Pact, k6, Playwright, Cypress, Jest, Pytest

AI Engineering / Developer Tools: harness engineering, MCP, Claude Code, Kiro, OpenCode CLI, prompt engineering

EXPERIENCE

Software Development Engineer

Jun 2025 – Dec 2025

Amazon

Seattle, WA

- Co-owned Amazon Seller Central's compliance platform, a business-critical system where UI regressions directly block seller listings and impact revenue.
- Designed and launched a **Java REST** BFF service for Amazon Seller Central's compliance platform, aggregating 3 internal APIs into a stable widget contract, centralizing orchestration/error handling, and reducing client round trips from 3 to 1 for revenue-critical seller listing workflows.
- Re-architected ActionWidget from hardcoded regulation-specific UI logic into a schema-driven platform using the Strangler Fig pattern; decoupled regulation logic from rendering so new compliance flows onboard via config alone with no code changes.
- Established **TDD** practices across the team by authoring tests around API contracts, error handling, and edge cases; enforced through **trunk-based development** and code review gates, stabilizing production releases and reducing regression risk on revenue-critical seller flows.
- Designed AI-assisted **Playwright** test authoring pipeline using **BDD** specifications as LLM context, cutting test authoring time 63%, raising E2E coverage 60%, and accelerating delivery velocity 30%.
- Raised code review quality by enforcing single-purpose diffs, separating structural refactors from behavioral changes; smaller PRs surfaced potential bugs earlier, reducing regression risk during an active critical migration.

Software Engineer

Aug 2024 – Jun 2025

Shoptaki

New York, NY

- Built a Node.js backend for a natural-language analytics interface that extracted live PostgreSQL schema metadata, relationships, constraints, and join patterns to generate context-aware SQL from user prompts.
- Added SQL validation, sanitization, and staging-database execution safeguards before query execution, reducing risk from malformed or unsafe LLM-generated SQL.
- Built the query interface in React and TypeScript to handle dynamically shaped result sets where columns and types were unknown at build time; implemented async execution states, error surfaces, and adaptive table rendering to handle unpredictable LLM output shapes.

Software Engineer

Feb 2021 – Jan 2023

CodeCraft Technologies

Bangalore, India

- Co-owned **React/TypeScript** component development for JioCinema on **Next.js 13** (Pages Router) with **Redux**; implemented **SSR** and code splitting via dynamic imports on streaming pages that served IPL and FIFA World Cup 2022 traffic, platform peak of 32M concurrent viewers.
- Built **Node.js** and Python backend services on **AWS** ECS/Lambda to process streaming-platform metadata and content workflows, exposing REST APIs and scheduled jobs for ingestion, transformation, and downstream UI consumption.
- Built full-stack audience-engagement features for Voot, including configurable surveys, show-specific quizzes, participant leaderboards, and real-time voting, deployed across multiple entertainment and reality-show properties on a platform serving 100M monthly active users.
- Modernized build pipeline for a high-traffic streaming platform (Webpack/Babel upgrade, asset optimization); cut build times 50% and unblocked ES6+ feature adoption.
- Migrated application state from **Redux** to **Apollo Client (GraphQL)**, implementing a unified GraphQL infrastructure layer; eliminated boilerplate and reduced bundle size 20%.

Assistant Systems Engineer

Nov 2018 – Apr 2020

Tata Consultancy Services

Bangalore, India

- Built real-time IoT monitoring dashboard using **Angular**, **Flask**, and **Server-Sent Events (SSE)**, handling continuous sensor data streams at scale; designed RESTful API endpoints for auth, data retrieval, and live updates.
- Optimized frontend performance via lazy loading, image compression, and JS minification; delivered measurable load time improvements on production dashboard.

EDUCATION

Pace University

Master of Science in Computer Science

Sri Siddhartha Institute of Technology

Bachelor of Engineering in Electronics and Communication Engineering

New York, NY

Jan 2023 – May 2024

Tumkur, India

September 2018